

NAME

ppqac - post-quantum access control: exoteric mediation, esoteric observation

SYNOPSIS

ppqac is a short paper.

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mandoc -T pdf ppqac.7 > ppqac.pdf
mandoc -T lint ppqac.7
```

ABSTRACT

Post-quantum *cryptography* is a claim about the residual hardness of named problems (lattices, hashes, codes) when the adversary's uniform class is *BQP* rather than *BPP*. Post-quantum *access control*, as used here, is a different claim: that the defender's *authorization* relation is not an instance of such a problem, and therefore does not collapse under Shor or Grover.

The defender is a classical stored-program machine (von Neumann: one linear store, code as data, binary words) running a FreeBSD kernel with mac(4) hooks. The adversary is quantum: polynomial-time quantum algorithms on the usual public-key names, Grover search over unstructured domains, and, as a working hypothesis, coupling to the electrical and electromagnetic plant (power distribution network, return, near-field) including entanglement or entanglement-enhanced metrology. The adversary is *not* assumed to be executing a quantum ISA on the defender.

Exoterically, discretionary access control, lattice mandatory access control, and MMU translation are interpreted predicates over credentials, labels, and address-space metadata. They do not reduce to discrete logarithm or period-finding. A BQP adversary who would *act as* a subject must still produce a classical write accepted by the vnode path, the MAC policy module, and the translation lookaside. What remains BQP-sensitive is *authentication* (binding a name to a subject before a label attaches): sshd(8), TLS, package signatures, then login.conf(5).

Esoterically, those predicates are themselves electrical events. A page-table walk is current on a rail; a label in a system extended attribute is a pattern of stored charge that, when interpreted, switches bitlines. The in-band/out-of-band cut is an alphabet cut (syscalls versus analog traces), not a cut in the physics. If an observation function on the plant has a kernel-visible subject in its factorization, the monitor still constrains the support of the observation. If it does not factor, the monitor is silent. Independence of the observation from high objects is *imperviousness* (a non-interference statement). A quantitative bound on leakage is *resistance*. These are not interchangeable. Quantum advantage on the plant is not a corollary of Shor. Packaging *A_M* on GENERIC (rc.d, setfsmac(8), ZFS, login classes, a 2001 PLM) supplied operational lemmas the slogans omit: empty lattices, first-match combinators, obsolete *Obj* alphabets, the login cut, RDTUN ifnets, and recovery as part of the monitor.

MACHINE AND ADVERSARY

Defender

A von Neumann host: shared store, sequential fetch, binary encoding. FreeBSD GENERIC provides ‘options MAC’. We take `mac_lomac(4)` as the integrity policy (floating labels, low-watermark) and Unix DAC as the residual access matrix on the same objects. The MMU implements a function from (virtual page, privilege, rights) to (frame, fault). Virtual memory may revoke `mmap(2)` mappings when LOMAC demotes a subject (`mac_lomac(4)`), *security.mac.mmap_revocation*. The TCB is the kernel plus the loaded policy modules. `mac(9)` does not claim complete root containment; neither do we.

This is not a quantum architecture. Qubit memory, CPTP evolution as the instruction semantics, and no-cloning as a confinement axiom are out of model. CHERI-style capabilities would be a different classical machine; they are not this paper.

Adversary

Uniform class *BQP* plus Grover’s quadratic over unstructured search (oracle) $O: \{0,1\}^n \rightarrow \{0,1\}$, expected $O(2^{\sqrt{n/2}})$ queries. Shor’s algorithm places integer factorization and discrete log in *BQP*; the usual PKI names (RSA, finite-field DH, ECDH, ECDSA) are therefore not authentication against this adversary. Grover does *not* yield a symbolic executor, a confused-deputy oracle, or a kernel-bug finder. Unstructured search over a 32-bit ASLR slide is incomparable to finding a `mac(9)` hook bypass.

Plant coupling (T3) is hypothesized, not demonstrated here: a completely-positive instrument on some electromagnetic degree of freedom correlated with defender computation, possibly entanglement-enhanced (spin squeezing, NV-center or SQUID magnetometry) beyond the standard quantum limit toward the Heisenberg limit. A 300 K CMOS die and copper PDN are a thermal bath; decoherence times for macroscopic entanglement with the rail are not assumed generous. T3 asserts only that some distinguishing observation may exist, not that it is cheap.

Out of scope

Replacing the naming path with ML-KEM, ML-DSA, or SLH-DSA is necessary and insufficient. Necessary: otherwise the adversary forges the name that `login(1)` will label. Insufficient: a correctly named subject is still mediated by DAC, MAC, and the MMU, which never consulted the modulus.

EXOTERIC: NAMING AND MEDIATION

Authentication binds an identity to a process. Authorization relates that process to objects and to regions of the store. The first, on contemporary FreeBSD, is cryptographic or a local console. The second is an access matrix (DAC: owner-directed) `chmod(2)` and `chown(2)`, `uid/gid` composed with a lattice policy (MAC) composed with translation rights (MMU).

Let *Sub* be subjects (credentials) and *Obj* objects (vnodes, mappings, sockets, ifnets). DAC is a function $A_D: Sub \times Obj \rightarrow 2^{\{ops\}}$ modifiable by the object’s owner. MAC is a function A_M computed from

labels *lambda(s)*, *lambda(o)* in a partial order (*L*, *preceq*) and not owner-modifiable. Anderson's reference monitor (complete mediation, isolation, verifiability) is approximated by *mac(4)* entry points on *vnode*, *VM*, *IPC*, and *network* objects. Complete mediation is an aspiration; *mac(9)* states that not all attack channels on a privileged subject are hooked.

Shor destroys *A_auth* when *A_auth* is RSA/ECC. It does not inhabit *A_D* or *A_M*. There is no reduction from period-finding to 'lomac/high' or to the allocation of a PTE. A BQP adversary who would become *s* in *Sub* must still induce a classical state the hooks accept. That is **quantum-adversary-stable mediation**: the decision procedure's complexity class is not the source of its security, because the decision is not a cryptographic game. It is an interpreted predicate in the kernel.

LATTICES

Bell-LaPadula is a secrecy lattice: no read up, no write down. Biba is an integrity lattice: no read down, no write up. *mac_lomac(4)* is Biba-like integrity with *subject relabeling* on read-down: a subject at grade *g* that reads an object at *g' < g* is demoted toward *g'* (low-watermark, floating label). Subsequent writes to objects dominated by the old *g* fail. Auxiliary grades '[aux]' and ranges '(lo-hi)' exist for exec-time assumption and for administrative exemption 'equal'.

This axiom is *not* non-interference for secrecy. Goguen-Meseguer non-interference, and the observational-determinism and unwinding lemmas used to prove it, speak to whether high inputs are visible in low outputs. LOMAC *permits* the read that secrecy forbids, then removes write-up. A side-channel thesis about emanations of high *secrets* is therefore not discharged by LOMAC; it would want MLS, Biba's no-read-down, or an enclave whose observation function is specified in analog. LOMAC's short proof, and the reason this suite uses it, is integrity: a sandbox subject cannot write-up into high OS objects after a low read.

setfsmac(8) composes specfiles by *first match* (later regexes for the same path are dropped silently). That is a policy combinator, not a lattice operation; overlays must precede the official PLM or a leading '/dev' catchall 'equal' rule shadows later high device nodes.

MECHANISM DIVERSITY

DAC, MAC, and MMU are not replicas of one lock (replication would share one specification and one failure). They are three interpretations of "may this subject disturb this region" with distinct failure modes:

- DAC fails when an owner (or a hijacked uid) grants *ops* the lattice would refuse.
- MAC fails when an object is unlabeled, when 'equal' is over-applied, when first-match drops a high rule, or when a hook is missing (*mac(9)*).
- MMU fails when a mapping is granted that the lattice would later wish revoked, when a wild pointer is already in-kernel, or when microarchitectural leakage (Spectre-class) reads across translations the PTE

still forbids architecturally.

A quantum adversary who factors one modulus walks around all three. A quantum adversary who obtains a classical write still faces the other two if the write is not already supervisor-mode. Diversity is not a work factor in n and does not grow with qubits. It is the absence of a single specification whose one bug is total.

Login remains the cut. `login.conf(5)` attaches *lambda(s)* after authentication. A forged name is a forged subject. Isolation after a correctly named login is what `mac_lomac_grok(8)` and `mac_grok(8)` implement: groups as roles, official contexts, ZFS recursive snapshots as recovery, PREINSTALL as behavioral restore. That practice is not argued here; the manuals argue it.

ESOTERIC: THE OBSERVATION FUNCTION

Extend the system with an observation alphabet disjoint from syscalls. Let Tr be executions (traces of hooks, translations, and pipeline switching) and let $O: Tr \rightarrow H$ map a trace into a Hilbert space (or, classically, into a time series in $\mathbb{R}^k$): supply current, chassis potential, near-field B , impedance echo on the adapter primary. TEMPEST, SPA/DPA, and RAMBO-style bus modulation are the classical O . T3 asserts that some quantum instrument may refine O .

In-band is the syscall alphabet. *Out-of-band* is O . Both are functions of the same electrons. A PTE walk is not adjacent to the PDN; it is a PDN event. A `chmod(2)` is charge in a line. The clerk's cut (kernel story versus physics) does not imply that authorization is "not electrical". It implies that the kernel names only some of the electrical events.

Say O *factors through a subject* when there exists a projection $\pi: Tr \rightarrow Sub$ (the kernel-visible principal whose switching dominated the analog) and a map $O': Sub \times Tr \rightarrow H$ with $O = O' \circ (\pi \times id)$ in the relevant coordinates. Then constraining π 's image (who may run, which objects they may map, whether they are demoted) constrains the support of O . Sandbox credentials, LOMAC demotion, seeotheruids, and page rights become statements about which traces are in the domain of a useful O' . That is resistance, if one then bounds $I(high; O)$ or an analogous distinguishing advantage.

Say O *does not factor* when an authorized hot core, or the idle plant, already yields a distinguishing observation of high objects with no unprivileged modulator. Then A_D , A_M , and the MMU are silent: the computation is allowed, and O reads it. The remaining claims are physical: constant-power dual-rail logic, filtered PDN, Faraday, spatial isolation of high work. Those do not follow from `mac_lomac.ko`.

Imperviousness is observational equivalence: for all traces t, t' agreeing on low inputs, $O(t)$ and $O(t')$ are indistinguishable (trace distance, or a computational distinguisher with the adversary's instrument). That is non-interference with respect to O , not with respect to `write(2)`. POSIX MAC does not supply it. LOMAC in particular supplies the wrong axiom for emanation *secrecy*: the authorized trusted subject computing on a high object is the hot core O wanted.

Cost of T3

Shor is a clean reduction in the quantum circuit model against a named algebraic problem. Coupling *O* to a multilayer board at several gigahertz, under a ground plane, in GENERIC's background switching, is an experimental physics problem. Entanglement with the rail must survive a thermal bath, or the enhancement must be in the *sensor* (squeezed spin, NV, SQUID) not in a Bell pair with the victim's electrons.

Placement, bandwidth, a correlator, and usually a local modulator remain. The modulator, if required, is a process, hence a candidate for *pi*. Do not take quantum advantage on the plant as a lemma from BQP.

THESES

Deniable separately.

T1

Membership in *BQP* does not by itself produce a classical write accepted by the reference monitor.

T2

A_D, *A_M*, and MMU translation are quantum-adversary-stable mediation on a von Neumann defender. They are post-quantum access control in that sense, and not post-quantum cryptography.

T3

(working hypothesis) Some instrument on the plant, possibly entanglement-enhanced, can distinguish defender computation.

T4

If *O* factors through a kernel-visible subject, T1 and T2 constrain *O*. If not, T1 and T2 are silent and the property is physical non-interference.

T5

Quantum advantage for *O* is not free. Decoherence, placement, and distinguishing a vnode in GENERIC's noise are the adversary's burden, not Shor's corollary.

T1 and T2 are implemented by this suite's rc.d packages. T3 is assumed so that T4 is utterable. T4 is two theorems until a coupling map is exhibited. T5 keeps T3 from swallowing T1 and T2. The following lemmas are residue from building those packages and from Handbook and mailing-list failures that the slogans do not mention.

PRAXIS**Empty interpretation**

Enabling *A_M* before *setfsmac(8)* is a different relation: unlabeled objects versus labeled processes, EPERM noise. *security.mac.lomac.enabled=0* until a *setfsmac(8)* probe and then *setfsmac(8)* is the statement that the lattice must be total on the mounted *Obj* before the monitor interprets it. *getfsmac(8)*

EINVAL until **mac_lomac** is loaded is a domain error, not a ZFS failure.

Combinators are not joins

setfsmac(8) first-match is not least upper bound in *L*. The official PLM lists /dev catchall 'equal' before /dev/mdctl 'high'; the latter never applies. Overlays (dev-modern, x11-xorg, site) precede the PLM because the combinator is regex order. A proof of *A_M* that quotes the PLM without quoting the combinator is a proof of a different policy.

The object alphabet moves

The 2001 PLM names agp(4), syscons, /tmp/.X11-unix. GENERIC now names vt(4), evdev(4) *event**, DRM *card** (master, treat as old agp, high) and *renderD** (unprivileged, equal), hid(4) *hidraw*. A policy written against obsolete names is an unlabeled hole. Handbook Xorg failure is often that hole, or mac_partition(4), not a mystery in the lattice.

Subjects are created at login

lambda(s) is a login.conf(5) field, compiled by cap_mkdb(1), attached by login(1). The shell used to edit the file is not in *Sub* under the new class. login.conf(5) has no group-to-label map; roles are pw(8) groups bound to classes. Expecting the editing session to be labeled is a type error.

Network objects are in the lattice

Untrusted ifnets default 'low'. *security.mac.lomac.trust_all_interfaces* is RDTUN (loader.conf(5)), not a live sysctl(8). Remote-only enforce demotes or kills the authentication path. *ptys_equal* covers the tty objects login(1) needs. Handbook: implement MAC over a remote connection with extreme caution. This is T1 applied to the administrator's own subject.

The store is not UFS-shaped

ZFS has no tuneufs(8) **multilabel**. Labels live in the system extattr namespace (OpenZFS) "freebsd:system" prefix. User-namespace 'xattr=sa' versus 'dir' is a different API; do not flip it to "fix" LOMAC. Probe setfsmac(8) on /tmp; abort with *enabled=0* if EINVAL persists after kldload(8). UFS 'multilabel' on / is not sticky without the Handbook *fstab* read-only dance.

Integrity is experienced as a bug

'_secure_path: unable to stat .login_conf' when a high subject uses su(1) into a low home is LOMAC/Biba as specified. 'whoami' prints 0 and su(1) says "who are you?" when *A_M* is unloaded while login.conf(5) still carries 'label=', or when *master.passwd* inherited a conflicting label. Root 'equal(equal-equal)' is exemption, not containment (mac(9)). Fraser's 'pump', 'syslogd', 'sshd' trusted-binary list is Linux LOMAC; this module uses 'equal' on /tmp, /var/log, /var/run, /sbin/dhclient 'high[low]', and *ptys_equal*. Do not copy the USENIX exception list into a specfile. Auxiliary grades may not apply to extattr I/O (PR 178667); do not rest a proof on aux covering xattrs.

Conjunction is not a product of proofs

mac_seeotheruids(4) is orthogonal (no labels) and a short proof. mac_ifoff(4) *other_enabled*=0 denies the ifnet, hence the authentication path. mac_partition(4) needs labels and fights Xorg and 'top'. Biba stacked on LOMAC is a product of incompatible axioms. ugidfw(8) empty is allow-all; new users miss rules until module reload. *specificgid* is one GID, historically argued both ways about root (kern/72263, bin/79714).

Recovery is inside the isolation claim

A lockout is a failed interpretation, not successful confinement. PREINSTALL restores *behavior*: unload, copy back login.conf(5) and loader.conf(5), cap_mkdb(1), restore login classes. MAC xattrs remain; A_M becomes the identity without the kld. zfs(8) **snapshot -r** (filesystems and zvols) is the dataset morphism. zpool-checkpoint(8) rewind discards later transaction groups and the checkpoint itself (one checkpoint per pool; OpenZFS 12646). **uninstall** is not wipe and not rewind. **test*** before each **one***.

Authentication still sits in front

None of the above places A_auth in a BQP-hard problem. T2 is mediation after a name. The suite does not kldload(8) Kyber.

The cook and O

If a plant observation needs a modulator, that process is in *Sub* and the sandbox, label, and mmap(2) revocation story applies (T4, factoring). If the trusted subject is already the hot core, LOMAC will not mute emanations of what it was allowed to compute (T4, non-factoring; change the lattice or specify analog O).

PROOF OBLIGATIONS

A proof of the esoteric half must:

1. Exhibit *O* (current, B-field, impedance, squeezed magnetometry) and the traces it distinguishes.
2. Factor or refuse to factor *O* through *pi*: *Tr* -> *Sub* (credential, mapping, vnode) versus the plant alone.
3. If it factors, show independence (imperviousness) or bound *I(high; O)* or a distinguisher's advantage (resistance). Do not conflate the two.
4. If it does not factor, discharge the remainder outside mac(4): PDN filter, constant power, shielding, air gap, or a secrecy lattice / enclave whose analog *O* is in the spec.
5. Preserve LOMAC's axiom. Do not derive no-read-down from a low-watermark. If the goal is emanation secrecy, say so and change the lattice.

SEE ALSO

cap_mkdb(1), chmod(1), login(1), mandoc(1), su(1), mmap(2), agp(4), evdev(4), hid(4), mac(4),

mac_ifoff(4), mac_lomac(4), mac_partition(4), mac_seeotheruids(4), vt(4), loader.conf(5), login.conf(5), mac_grok(7), mac_lomac_grok(7), getfmac(8), kldload(8), mac_generic_grok(8), mac_grok(8), mac_lomac_grok(8), pw(8), setfmac(8), setfsmac(8), sshd(8), sysctl(8), tuneefs(8), ugidfw(8), zfs(8), zpool-checkpoint(8), mac(9)

Anderson, J. P., "Computer Security Technology Planning Study", ESD-TR-73-51, 1972 (reference monitor). Bell, D. E. and LaPadula, L. J., ESD-TR-73-278 (secrecy lattice). Biba, K. J., "Integrity Considerations for Secure Computer Systems", MTR-3153, 1977. Fraser, T., "LOMAC: Low Water-Mark Integrity Protection for COTS Environments", USENIX FREEENIX 2001. Goguen, J. A. and Meseguer, J., "Security Policies and Security Models", IEEE S&P 1982 (non-interference). Grover, L. K., STOC 1996. Harrison, M. A., Ruzzo, W. L., Ullman, J. D., "Protection in Operating Systems", CACM 19(8), 1976. Shor, P. W., FOCS 1994. NIST PQC, for *A_auth* only. The FreeBSD Handbook ch.19 (MAC practice and troubleshooting).

HISTORY

Spoken as a refinement of "post-quantum access control" on a classical mac_lomac(4) host: not a cryptographic suite, not a quantum OS, a BQP adversary, then the plant, then the observation function. Rewritten for a technical audience, then extended with praxis from packaging *A_M* on GENERIC the same day.

AUTHORS

Grok and the operator of **freebsd-mac-grok**, 2026-08-22. T1 and T2 are the packages. T3 and T4 remain the operator's.

BUGS

"Post-quantum" denotes Kyber unless **pqac** is said aloud. The defender's ISA is binary stored-program, not a quantum ISA. Grover is not an exploit generator. T3 is neither free nor forbidden here. kldload(8) of **mac_lomac** is neither Faraday nor a naming retrofit; it is a dominance test on a labeled store. This paper is unsigned; a later PGP pass is out of band.